

Generalized Pigeonhole Principle

Red Versus Blue — Understanding the Intuition Using the Generalized Pigeonhole Principle

This problem is one of the cleanest examples of applying the **Generalized Pigeonhole Principle** in a constructive algorithm.

The key observation is that the problem is **not really about red and blue blocks**. It's actually about **distributing objects as evenly as possible**.

Let's build the intuition from scratch.

Problem Statement

You are given:

- R red blocks
- B blue blocks

Your task is to arrange all the blocks so that the **maximum number of consecutive red blocks is as small as possible**.

For example,

```
R = 8
B = 2
```

How should we arrange them?

Step 1: Ignore the Red Blocks

Instead of thinking about both colors together, first ignore the red blocks completely.

Place only the blue blocks.

```
B  B
```

Now ask yourself:

Where can the red blocks go?

They can only be placed

- before the first blue,
- between two blue blocks,
- after the last blue.

So the arrangement actually looks like

```
_ B _ B _
```

The underscores represent places where red blocks can be inserted.

These empty spaces are called **gaps**.

Step 2: Count the Number of Gaps

Suppose there are B blue blocks.

```
B = 1
_ B _
2 gaps
```

```
B = 2
_ B _ B _
3 gaps
```

```
B = 3
_ B _ B _ B _
4 gaps
```

A pattern appears.

If there are B blue blocks, they always create $B + 1$ gaps.

This is the most important observation in the entire problem.

The blue blocks are **not the main focus**.

Their only purpose is to create containers where the red blocks can be placed.

Step 3: Convert the Problem into a Distribution Problem

Now forget the colors again.

We now have

- R red blocks
- $B + 1$ gaps

Think of it this way:

- Red blocks = **Objects (Pigeons)**
- Gaps = **Boxes (Pigeonholes)**

So the original problem becomes

Distribute R identical objects into $B + 1$ boxes so that the largest box is as small as possible.

Once you reach this point, the colors no longer matter.

This is now a classic **Generalized Pigeonhole Principle** problem.

Step 4: Apply the Generalized Pigeonhole Principle

The principle tells us:

If m objects are distributed among n boxes, then at least one box must contain

$$\left\lceil \frac{m}{n} \right\rceil$$

objects.

In our problem,

- Objects = R
- Boxes = $B + 1$

Therefore,

the largest gap **must contain at least**

$$\left\lceil \frac{R}{B + 1} \right\rceil$$

red blocks.

This value is unavoidable.

No arrangement can do better.

Why?

Suppose we try to make every gap smaller than

$$\left\lceil \frac{R}{B + 1} \right\rceil$$

Then the total number of red blocks would become too small.

So it becomes mathematically impossible to place all the red blocks.

That is exactly what the pigeonhole principle proves.

Example 1

```
R = 8
B = 2
```

Blue blocks create

```
_ B _ B _
```

Number of gaps

```
3
```

Now distribute

```
8 reds
```

among

```
3 gaps
```

Average reds per gap

```
8 / 3 = 2.67
```

Since we cannot split a red block,
the largest gap must contain

```
ceil(8/3) = 3
```

reds.

Can every gap contain only 2 reds?

Let's see.

```
2 + 2 + 2 = 6
```

But we have

```
8 reds
```

There are still

```
2 reds
```

left to place.

So at least one gap must contain **3** reds.

This is unavoidable.

One Optimal Arrangement

```
RRR B RRR B RR
```

Gap sizes are

```
3
3
2
```

Maximum consecutive reds

```
3
```

And that equals

```
ceil(8/3)
```

So this arrangement is optimal.

Example 2

```
R = 10
B = 3
```

Blue blocks create

```
_ B _ B _ B _
```

Number of gaps

```
4
```

Average

```
10 / 4 = 2.5
```

Therefore,

```
ceil(10/4) = 3
```

The best distribution is

```
3
3
2
2
```

Arrangement

```
RRR B RRR B RR B RR
```

Maximum run

```
3
```

Again,

this equals

```
ceil(10/4)
```

Example 3

```
R = 11
B = 3
```

Blue blocks create

```
4 gaps
```

Now distribute

```
11 reds
```

among

```
4 gaps
```

Average

```
11 / 4 = 2.75
```

Largest gap

```
ceil(2.75) = 3
```

One balanced distribution is

```
3
3
3
2
```

Maximum run

3

Again, this is optimal.

Why Does the Codeforces Solution Use Division and Remainder?

Almost every solution computes

```
C++
base = R / (B + 1);
extra = R % (B + 1);
```

Let's understand why.

Suppose

```
R = 11
B = 3
```

Number of gaps

4

First,

give every gap the minimum equal share.

$$11 / 4 = 2$$

So every gap gets

```
2
2
2
2
```

Total reds used

$$2 \times 4 = 8$$

Remaining reds

$$11 - 8 = 3$$

Now distribute these remaining reds.

Instead of putting all three into one gap,

place **one extra red into different gaps**.

```
3
3
3
2
```

Notice something beautiful.

The difference between any two gaps is never more than 1.

That means the distribution is as balanced as mathematically possible.

This is exactly why the code does

```
C++
for(each gap)
{
    put base reds;

    if(extra > 0)
    {
        put one more red;
        extra--;
    }
}
```

Every gap starts with the same number of reds.

Only the first **extra** gaps receive one additional red.

This guarantees the smallest possible maximum consecutive reds.

Why Does This Always Work?

Suppose two gaps differ by more than one.

For example,

```
5
2
```

One gap has

```
5
```

reds,

another has

```
2
```

reds.

Clearly,

we can move one red

```
5 → 4
2 → 3
```

Now the largest gap becomes smaller,

which means the previous arrangement was not optimal.

Therefore,

in the optimal arrangement,

the sizes of any two gaps differ by **at most 1**.

This is exactly what division and remainder achieve.

The Real Intuition

The problem is **not really about arranging colors**.

It is secretly asking:

How do we distribute R identical objects into $B + 1$ containers so that the largest container is as small as possible?

The blue blocks simply create the containers.

The red blocks are the objects we must distribute.

Once you recognize this transformation, the entire problem becomes a direct application of the **Generalized Pigeonhole Principle**.

How to Recognize This Pattern in Future Problems

Whenever a problem asks you to:

- minimize the largest group,
- split items into segments,
- distribute work evenly,
- balance loads,
- partition elements,
- minimize the maximum size of any partition,

ask yourself two simple questions:

1. What are the objects?

These are the things being distributed.

Examples:

- tasks,
- balls,
- people,

- books,
- red blocks,
- jobs.

2. What are the boxes?

These are the containers created naturally by the problem.

Examples:

- workers,
- shelves,
- groups,
- partitions,
- gaps,
- machines.

If you can identify the objects and the boxes, you've often uncovered a **Generalized Pigeonhole Principle** problem.

Applying This to "Red Versus Blue"

Problem Element	Interpretation
Red blocks	Objects (Pigeons)
Blue blocks	Create separators
Gaps between blue blocks	Boxes (Pigeonholes)
Number of gaps	$B + 1$
Goal	Minimize the largest gap
Smallest possible largest gap	$\left\lceil \frac{R}{B+1} \right\rceil$
Construction	Give every gap $R / (B+1)$ reds, then distribute the remainder one by one

Final Takeaway

The biggest lesson from this problem is the **transformation of perspective**.

At first glance, it looks like an arrangement problem involving colors.

But the moment you realize:

- **Blue blocks only create gaps**, and
- **Red blocks are simply being distributed among those gaps**,

the problem transforms into a classic distribution problem.

From there, the Generalized Pigeonhole Principle immediately tells you the optimal answer:

$$\text{Minimum possible maximum consecutive reds} = \left\lceil \frac{R}{B+1} \right\rceil$$

And the implementation naturally follows by distributing the reds as evenly as possible using **quotient (base) and remainder (extra)**. This is why this Codeforces problem is considered a textbook application of the generalized pigeonhole principle.

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